

DALAS Project

Drug Repurposing

Vladislav Antipin (21242244)

Paul Béglin (21408798)

Master of Computer Science, Sorbonne University

January 9, 2026

Contents

1	Introduction	3
1.1	Problem Statement and Objectives	3
1.2	Biological Rationale: The Drug–Pathway–Disease Network	3
2	Data	4
2.1	Data Sources	4
2.2	Target variable definition	4
2.3	Feature Engineering and Description	5
2.4	Exploratory Data Analysis	6
2.4.1	Missing Values	6
2.4.2	Univariate and Bivariate Analysis	7
2.4.3	Dimension Reduction and Outlier Analysis	8
2.4.4	Supervised Dimension Reduction	9
2.5	Graph Representation	10
3	Methods	10
3.1	Model Overview	10
3.1.1	Linear Models	10
3.1.2	Tree-Based Ensembles	11
3.1.3	Graph-Based Model	11
3.2	Implementation Details	11
3.3	Evaluation Strategy	12
4	Results	12
4.1	Model Comparison	12
4.2	Feature Analysis	13
4.3	Failure and Success Cases	14
5	Discussion	15
6	Conclusion	16

A	Appendix	18
A.1	Supplementary Tables	18
A.2	Supplementary Figures	19
A.3	Supplementary Algorithms and Listings	30

1 Introduction

1.1 Problem Statement and Objectives

The discovery of new drugs is crucial but costly and time-consuming, and many candidates ultimately turn out to be ineffective or toxic. It is thus of interest to repurpose already approved drugs for diseases they have not yet been tested against. To prioritize potential clinical trials, in addition to experts’ knowledge, it’s important to be able to predict the likelihood of success for such studies. Predicting drug repurposing success has both academic and clinical value, as it helps reduce experimental costs and accelerates the discovery of effective treatments.

The main objectives of this project are to explore public datasets for relevant features, develop and evaluate a preliminary machine learning pipeline for success prediction, and identify biases and potential improvements. Unlike industrial or academic approaches, this work is exploratory and focused on initial modeling.

We decided to specify a group of diverse but closely related diseases - immune system disorders, and drugs that are approved for at least one member of this group. Figure 1 summarizes pipeline, which will be presented in detail in the following sections.

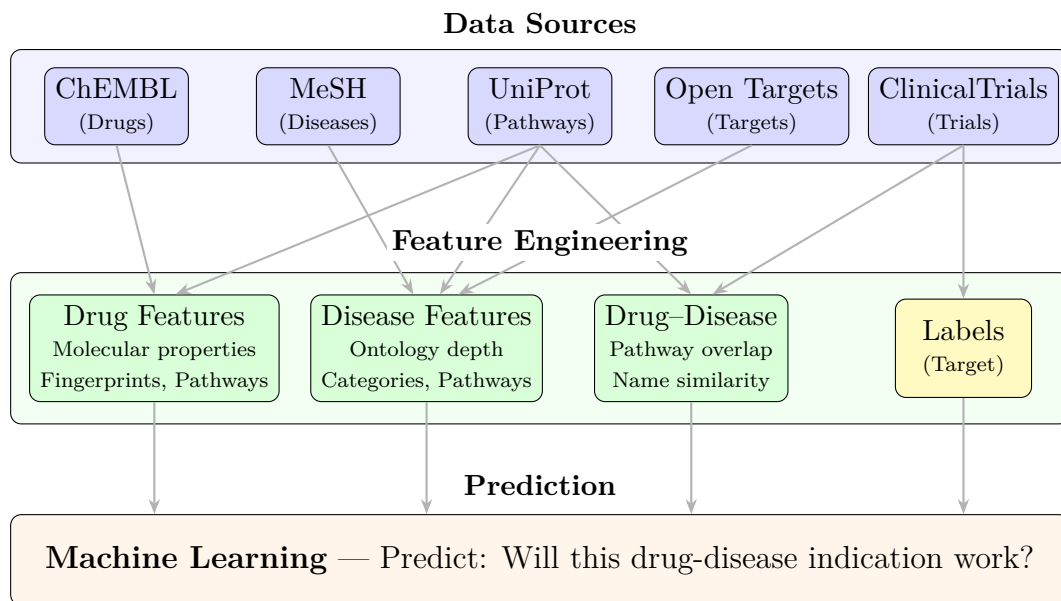


Figure 1: Complete pipeline overview for drug repurposing prediction.

1.2 Biological Rationale: The Drug-Pathway-Disease Network

One of the key assumptions underlying drug repurposing is that drugs and diseases are connected through shared biological pathways. Figure A1 illustrates this relationship as a tripartite graph. This structure motivates our `path_similarity` feature: we compute the Jaccard similarity between the set of pathways associated with a drug and those associated with a disease. High similarity suggests the drug affects biological processes relevant to the disease, making it a stronger repurposing candidate. Pathway-mediated connections can provide mechanistic justification for predictions. However it is important to highlight that this approach is extremely simplified and doesn’t capture the directionality of interactions (e.g. positive vs. negative regulation) as well as complex interactions

and feedback loops.

2 Data

2.1 Data Sources

We focused our attention on a few public databases having information on basic classification of diseases (MeSH), drug chemical descriptors and targets (ChEMBL), disease targets (Open Targets) as well as information on drug–disease indication studies (ClinicalTrials). Table 1 summarizes information on these data sources. Although initially we were using the web scraping to get information on disease classification, we found out that this way of getting medical information is rather deprecated, so we queried the corresponding databases instead. Data concerning drugs and diseases were combined using the cross references furnished by the same databases (ChEMBL and Open Targets). We also used a UniProt identifier mapper that allowed us to unify synonymous target IDs as well as enrich them with corresponding molecular pathway identifiers from Reactome database.

Table 1: Description of data sources.

Source	Access Method	Data Retrieved
MeSH RDF	SPARQL endpoint	Disease IDs and ontology
ChEMBL	REST API (Python client with query-style filters)	Drug chemical properties, targets and indications
UniProt	REST API (Python client)	Map target and pathway IDs
Open Targets	GraphQL API	Disease targets
ClinicalTrials.gov	REST API	Trial results and metadata

2.2 Target variable definition

The prediction target is a success–fail variable indicating whether a drug shows an effect against a disease; this constitutes a binary label for each drug–disease indication. Drugs that passed the fourth (final) clinical trial phase for a given disease are labeled as successful. For other cases, the outcome is less clear: if the last phase wasn’t (yet) passed, we can understand the reason by addressing the corresponding clinical trials. There are several possibilities: the combination was never tried (most common), was tried but ineffective, was effective only in earlier phases, or has no published results due to study termination or other reasons. There exists some manually and automatically annotated clinical trial databases [Gao et al., 2024] [Chen et al., 2025], but they are either limited in the time span (e.g. last 10 years), or their "non-success" label simultaneously encompasses trials failed due to drug inefficiency and other reasons. Therefore, we decided to automatically label drug–disease indications that have been tried at least once using rules detailed in Algorithm A1 (the complete algorithm can be found in the source code in the data preparation section).

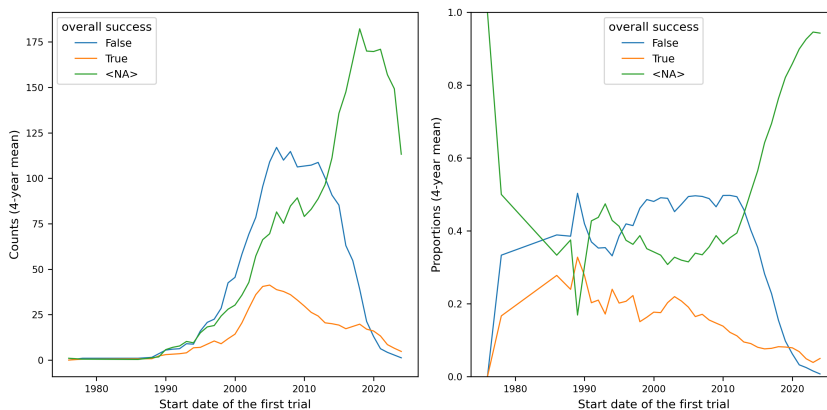


Figure 2: Counts and proportions of drug–disease indications by success label over time.

Interestingly, most undecided indications and clinical trials without published results occur roughly before 2000 and after 2010, as shown in Figures 2 and A2. One can speculate that it is linked to the fact that ClinicalTrials.gov was launched in 2000 and it became mandatory to publish there all studies (but not full results) since 2007. As can be seen in Figure A3, among indications with no first trial start date available, most are labeled successful, which mostly corresponds to indications established long before the launch of ClinicalTrials.gov. It’s important to note that some of those choices are arbitrary, which likely constitutes the most significant limitation.

2.3 Feature Engineering and Description

Detailed description of extracted features can be found in Table A1. Most drug features describe chemical and structural properties. Among engineered features are Morgan molecular fingerprints with 2 atoms radius obtained with RDKit module from drug’s SMILES (flat string representation of the structure) and yielding a 2048 dimensional bit vector. Each bit of this vector encodes the presence or absence of a particular sub-structure. To reduce the dimensionality, first 100 dimensions were taken and labeled as `FP_#`.

Disease features consist of disease ontology description, one of them being ontological category in addition to "immune system diseases", labeled as `cat_#`, for example some of immune diseases are as well neoplastic (e.g. leukemia).

For both diseases and drugs, we obtained a list of molecular targets, focusing on proteins. Since targets themselves are highly heterogenous and deeply interconnected within molecular pathways, we converted each drug’s or disease’s target list into a corresponding list of pathways. We then applied TF-IDF vectorizations separately to the pathway lists for drugs and for diseases. This approach downweights common housekeeping pathways while highlighting more informative ones. We then performed PCA and extracted first 50 principal components for drugs and 50 for diseases, labeled as `drug_path_#` and `disease_path_#`. This information captures drug-drug and disease-disease mechanistic similarity based on shared pathways.

To capture the drug–disease pathway similarity, we computed a Jaccard similarity index based on corresponding pathway ID lists. We also considered including drug–disease name similarity based on PubMedBERT embeddings, which reflect how closely a drug and a disease are associated in the scientific literature. However, we later realized

that including this feature in model training could introduce a data leakage risk, since PubMedBERT is trained on recent literature.

2.4 Exploratory Data Analysis

2.4.1 Missing Values

Among collected data and selected features, only a specific group of drug features contains missing values (Figure 3). To investigate the cause, we created an indicator variable for missingness and visualized patterns with a correlation heatmap (Figure 4). Indeed, this group of chemical descriptors (such as number of aromatic rings or lipophilicity) tends to be absent together and is associated with biotherapeutics marker. Biotherapeutics are drugs derived from living organisms rather than being chemically synthesized. Most of them are antibodies or other proteins - large biopolymers for which these chemical descriptors are not relevant.

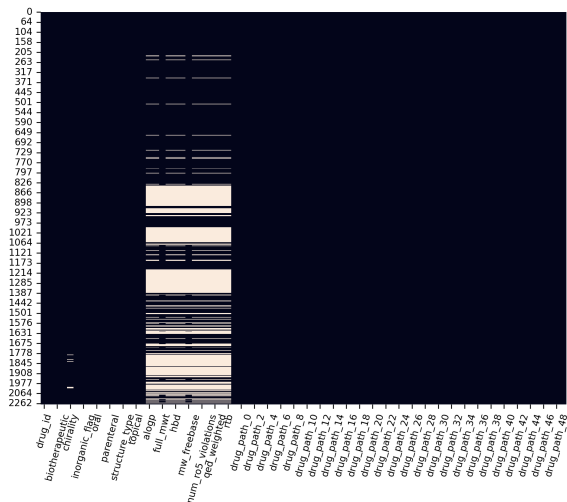


Figure 3: Missing values heatmap in the drugs dataset.

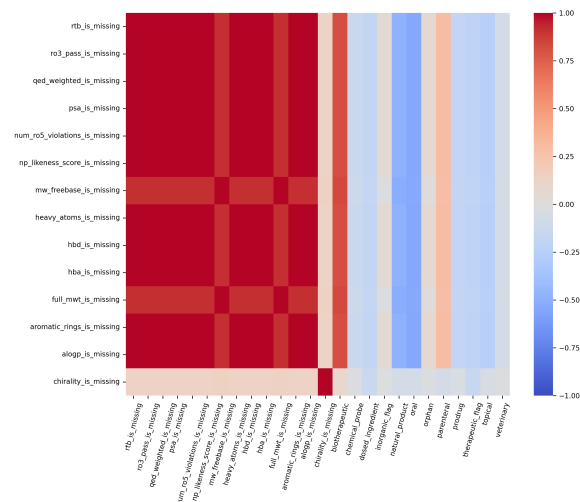


Figure 4: Missingness analysis of drug descriptors with correlation heatmap.

Inspection of drugs with missing values confirms this (e.g. insulin appears among such drugs). Therefore, we restricted the analysis to small-molecule drugs, since chemical descriptors are not applicable to biopolymers and imputing them would be inappropriate. After removing these cases, the data was ready for merging, resulting in a dataset containing complete information for each drug–disease indication. Table 2 provides a summary of data dimensions and label counts.

Table 2: Summary of data dimensions and label counts.

Dataset	Rows	Columns
Drugs	525	181
Diseases	149	79
Trials	14316	12
Indications	2482	12
Name embeddings	489939	5
Molecular fingerprints	1029	101
Merged data	890	252
Merged data success counts	False: 637, True: 253	

2.4.2 Univariate and Bivariate Analysis

Inspection of feature distributions across label values for variables unrelated to molecular pathways or fingerprints (Figure A4) shows mostly bell-shaped distributions, sometimes multimodal or slightly skewed. Log transformation was therefore considered unnecessary. Some disease ontology variables and drug descriptors exhibit outliers, which likely reflect normal biological or chemical variability and were further examined with dimension reduction techniques. Overall, feature distributions are similar between success and failure groups, as expected given the limited discriminative power of basic descriptors alone. However, engineered name and path similarity features tend to be higher for successful indications, suggesting potential predictive value when combined with other variables. Name similarity showed similar trend but was excluded from further analysis due to potential information leakage.

On a correlation heatmap of the full merged dataset (Figure A5), reveals, as expected, clusters of strongly or moderately associated variables. Notably between chemical descriptors, molecular fingerprints, and drug pathway embeddings, as well as between disease ontology types and disease pathway embeddings.

Focusing on more interpretable features only (Figure 5), allows straightforward interpretation of some associations. For instance, the number of heavy atoms is associated with molecular weight. Number hydrogen bond acceptors and donors (HBA and HBD) correlate positively with with polar surface area (PSA) and negatively with octanol-water partition coefficient (AlogP), reflecting increased hydrophilicity. Number of Lipinski rule-of-five violations is consistently associated with these descriptors. Interestingly, the number of aromatic rings correlates negatively with the natural product-likeness score (NP-likeness score), suggesting that purely synthetic drugs tend to contain more aromatic rings. Regarding disease ontology, immune diseases labelled as hemic or lymphatic tend to be classified as neoplastic, which is consistent with the fact that most immune-related cancers are leukemias or lymphomas. Another example is the positive association between urogenital diseases and infections.

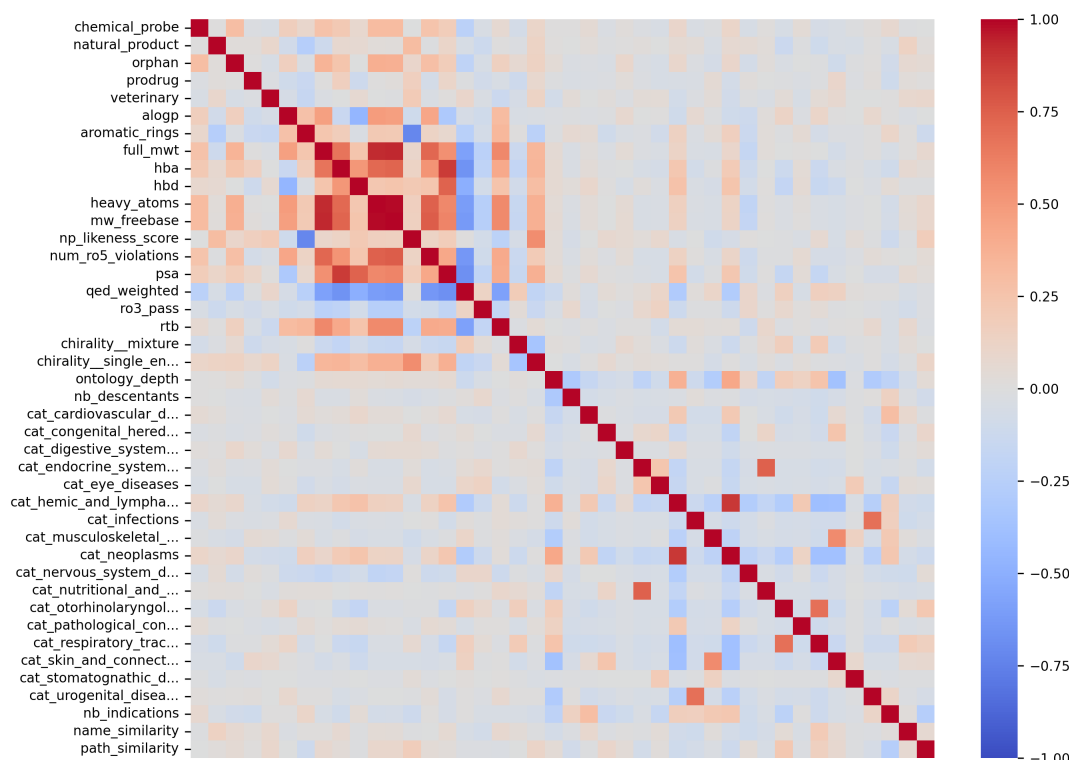


Figure 5: Focused view of the correlation heatmap on interpretable features.

2.4.3 Dimension Reduction and Outlier Analysis

PCA performed on centered and scaled merged data (Figure 6) shows that first two principal components explain only about 10 % of total variance, with variability spread across many components, as indicated by the scree plot (Figure A6). This suggests the absence of dominant linear axes of variation, consistent with an intrinsically high-dimensional feature space and likely non-linear associations in such a complex system. Furthermore, many features were explicitly constructed to be uncorrelated, such as principal components derived from molecular fingerprints and pathway embeddings. Nevertheless, inspection of the projection onto the first two components reveals that the main source of variation is driven by disease heterogeneity. In particular, most influential variables in the correlation circle (Figure 7) correspond to disease category indicators and disease pathway embeddings, distinguishing, for example, neoplastic blood disorders from respiratory tract diseases.

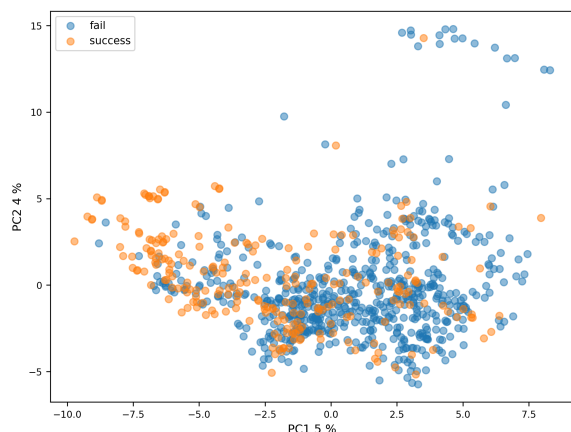


Figure 6: PCA projection of samples onto the first two principal components.

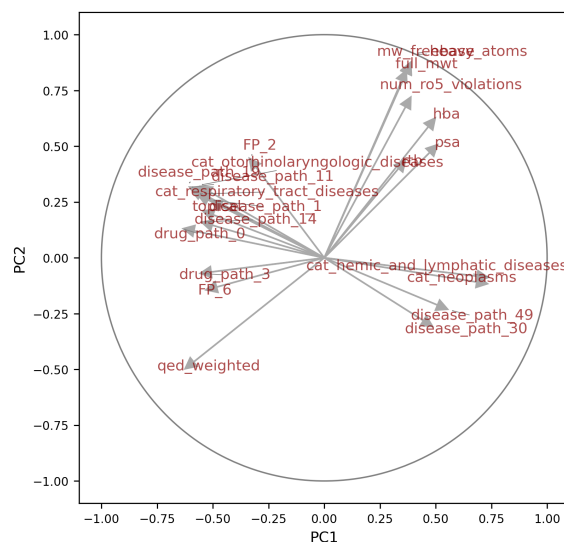


Figure 7: PCA correlation circle showing feature contributions to the first two principal components. Only arrows with length above 0.5 are shown.

The second principal component is primarily associated with drug-related variability. In particular, a small group of potential outliers appears at high values of both first and second principal components. Closer inspection reveals that these indications involve three drugs - sirolimus, everolimus and ridaforolimus, which are antineoplastic and immunosuppressive drugs sharing a common structural core - large macrolide ring (Figure A7). We therefore interpret these observations as a part of normal chemical variability rather than errors or inadequate filtering and chose not to exclude them.

Although PCA is an unsupervised method, a posteriori labeling reveals a slight separation between success and failure cases along the diagonal defined by the first two principal components. Notably, the Quantitative Estimate of Druglikeness (QED) contributes to both components and is aligned with successful indications. This suggests that the dominant directions of variability capture, at least partially, differences related to indication success. This pattern may also reflect differences in success rates across disease types; for instance, clinical studies in neoplastic blood diseases may exhibit lower apparent success rates due to higher study frequency or greater clinical difficulty.

Non-linear dimension reduction methods such as UMAP and t-SNE (Figures A9 and A10) reveal a large number of small clusters. These clusters appear to correspond to the indications involving identical or closely related drugs or diseases. In contrast, Isomap (Figure A8) does not reveal any easily interpretable structure in the data.

2.4.4 Supervised Dimension Reduction

Supervised dimension reduction using LDA and PLS-DA (Figures 8 and 9) shows that, although overlapping, success and failure labels are relatively well separated along the LDA axis and in projection onto the first two PLS-DA components.

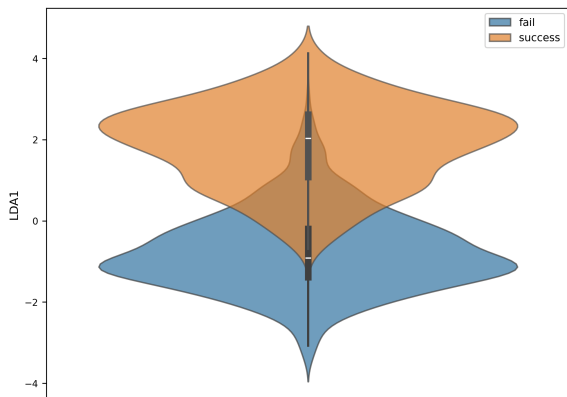


Figure 8: Linear Discriminant Analysis (LDA) projection highlighting class separation.

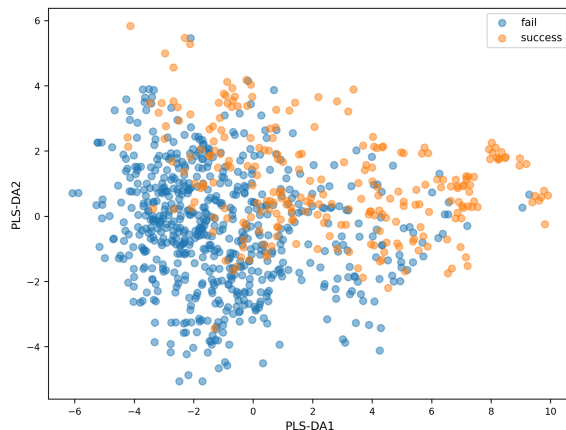


Figure 9: Partial Least Squares Discriminant Analysis (PLS-DA) projection of samples highlighting class separation.

2.5 Graph Representation

Although our binary classification approach for drug–disease indications is valid, it does not fully capture the true structure of the problem. The data are inherently structured as a bipartite drug–disease graph, with connections mediated by associated molecular pathways. Most advanced drug discovery research leverages molecular interaction networks and Graph Neural Networks directly, rather than relying on manually engineered features [Keramida et al., 2025]. To explore this, we organized the same data as a **HeteroData** graph using the PyTorch Geometric module (Listing 1). Building a robust graph-based model would likely require richer information about molecular and mechanistic connections, as well as a larger number of observations.

3 Methods

3.1 Model Overview

Since the prediction target is a binary success/failure for given drug–disease indication, we evaluated several supervised models for binary classification, including Logistic Regression, k-Nearest Neighbors (kNN), Support Vector Machines (SVM), Random Forest, Gradient Boosting, and XGBoost.

3.1.1 Linear Models

Logistic regression estimates probability of success using sigmoid-transformed linear combinations of the input features. Its biggest advantage is high interpretability feature weights indicate their importance, however it is limited to linearly separable problems.

k-Nearest Neighbors is an instance-based model, making predictions by based labels of the closest points in the feature space. Therefore, there is no actual training phase and the assumption behind this model is that similar observations tend to have the same label. Nevertheless, it is very sensitive to noise and the curse of dimensionality, as well as poorly interpretable (only by samples that voted).

Support Vector Machines fit a decision boundary that maximizes the margin between classes. Although it may be similar to regularized linear regression, the use of kernel functions allows to fit non-linear decision boundaries.

3.1.2 Tree-Based Ensembles

Tree-based ensemble models leverage hierarchical decision rules and combine the predictions of multiple models. Random Forest aggregates multiple decision trees trained on random subsets of variables and observations, which reduces the overfitting characteristic of individual decision trees. Gradient boosting and XGBoost, in contrast, build trees sequentially, with each new tree learning errors of the previous ones. XGboost is an optimized version of Gradient Boosting, offering additional parameters for fine-tuning.

3.1.3 Graph-Based Model

For graph-based approach, we used a Heterogenous Graph Neural Network (GNN) link classifier from PyTorch Geometric module. Unlike models that learn patterns based on tabular features only, this GNN learns representations by propagating information across connected nodes and edges, allowing it to capture more complex relationships. While this approach avoids manual feature engineering, it requires a large number of samples and rich, informative graph connections, and, like other neural networks, is prone to overfitting.

3.2 Implementation Details

All numeric features were centered and scaled, all categorical variable were one-hot encoded. No imputation was necessary, as observations with missing values corresponding to biopolymer drugs were excluded.

Most of the mentioned models are imported from scikit-learn module, except for XGBoost imported from `xgboost` and GNN from `torch_geometric` modules. Hyperparameter tuning was performed using a grid-search cross-validation `GridSearchCV` with search the spaces shown in Table 3.

Since most of the clinical trials, and therefore corresponding indications, tend to fail, the classes are imbalanced, with failures approximately three times more frequent than successes in our data. Class balancing options were applied where available (all models except for kNN and Gradient Boosting).

Table 3: Hyperparameter search spaces for each classifier.

Model	Hyperparameters
k-Nearest Neighbors	$n_neighbors \in \{5, 10, 20, 30\}$, $weights \in \{\text{uniform, distance}\}$, $p \in \{1, 2\}$
Logistic Regression	$C \in \{0.01, 0.1, 1, 10, 100\}$, $l1_ratio \in \{0, 0.25, 0.5, 0.75, 1\}$
SVM (linear)	$C \in \{0.01, 0.1, 1, 10\}$
SVM (polynomial)	$C \in \{0.1, 1, 10\}$, $degree \in \{2, 3, 4\}$, $gamma \in \{\text{scale, auto}\}$
SVM (RBF)	$C \in \{0.1, 1, 10\}$, $gamma \in \{\text{scale, auto}\}$
Random Forest	$n_estimators \in \{100, 300, 500\}$, $max_depth \in \{\text{None}, 10, 20\}$, $min_samples_split \in \{2, 5\}$, $min_samples_leaf \in \{1, 2\}$
Gradient Boosting	$n_estimators \in \{100, 300\}$, $max_depth \in \{3, 5\}$, $learning_rate \in \{0.01, 0.05, 0.1\}$, $subsample \in \{0.8, 1.0\}$
XGBoost	$n_estimators \in \{100, 300\}$, $max_depth \in \{3, 5\}$, $learning_rate \in \{0.01, 0.05, 0.1\}$, $subsample \in \{0.8, 1.0\}$, $colsample_bytree \in \{0.8, 1.0\}$, $gamma \in \{0, 0.1\}$

3.3 Evaluation Strategy

Prediction of drug–disease indication success requires avoiding temporal data leakage. To simulate a realistic scenario, we used older research results to predict success of new indications. Accordingly, the train-test split was based on the start date of the first clinical trial for each indication, with 15% of the data allocated to the test set, corresponding to a temporal cutoff in August 2014.

The training set was further used for hyperparameter tuning via grid search with 5-fold cross-validation, and model selection was based on cross-validation performance.

Because classes are imbalanced, accuracy is not an appropriate evaluation metric, as it would underemphasize errors on the minority (success) class. Therefore, we used F1 score (the harmonic mean of precision and recall) for model hyperparameter tuning, model selection and evaluation. Additionally, precision-recall curves were employed instead of ROC curves to analyze how classification performance varies with different decision thresholds (applicable only to models that output class probabilities).

4 Results

4.1 Model Comparison

The best estimators after hyperparameter tuning are presented in Table 4, with their corresponding hyperparameters in Table A2. Based on mean cross-validation F1 scores, all models perform relatively similarly, ranging from 0.60 to 0.65. XGBoost achieved the highest performance, while kNN performed the worst, which is expected given high dimensionality of the problem.

Table 4: Model performance after hyperparameter tuning (5-fold cross validation).

Model	Mean F1 score	Std F1 score
XGBoost	0.656383	0.038165
Logistic Regression	0.619974	0.037465
Support Vector Machine	0.619910	0.033970
Gradient Boost	0.609287	0.092787
Random Forest	0.601969	0.044340
k-Nearest Neighbors	0.591973	0.052311

We then used XGBoost for final evaluation on test, achieving a precision of 0.658, recall of 0.595 and F1 score of 0.625 (Figure 10).

The precision-recall curve (Figure 11) shows a rather smooth decline from 1 to chance level (success prevalence of 0.313), indicating that precision does not drop abruptly as the model becomes less strict (higher recall) and that overall performance is relatively stable. The area under the ROC curve is 0.824 (Figure A11), however, this metric is less reliable in the presence of class imbalance.

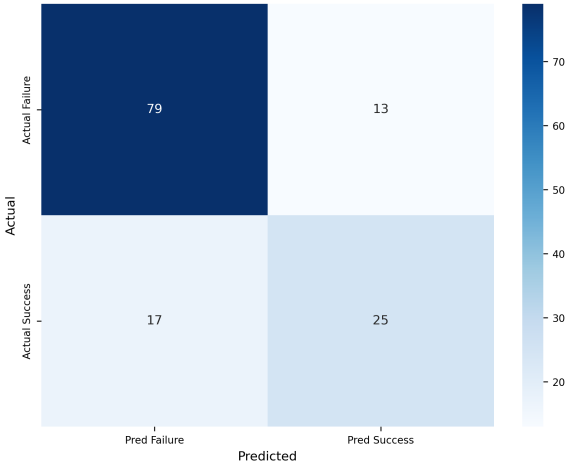


Figure 10: Confusion matrix for XGBoost on the test dataset.

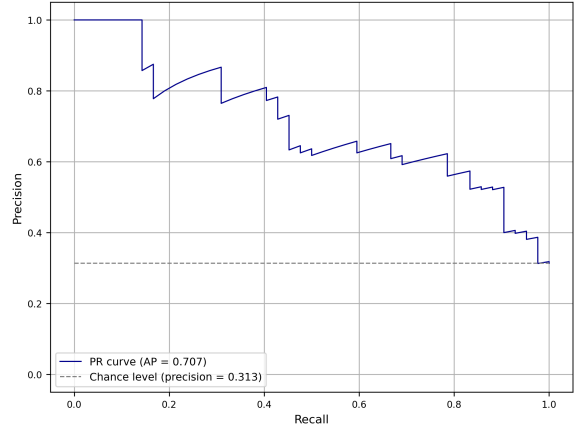


Figure 11: Precision-recall curve for XGBoost on the test dataset.

We also trained a Heterogenous GNN, achieving a precision of 0.429, recall of 0.714 and F1 score 0.535 (Figure A12). Unlike the other machine learning models, the GNN was evaluated independently for two reasons. First, GNN implementation in this project is rather exploratory, and we cannot yet ensure that all learning parameters were correctly configured. Second, differences in model architecture make direct comparison with other models in the standard cross-validation setting non-trivial. Therefore, GNN results should be interpreted as preliminary and complementary.

4.2 Feature Analysis

Since XGBoost model is not interpretable by design, we applied post-hoc explanation methods. We used permutation feature importance (evaluated using the F1 score) and

partial dependence analysis from `sklearn` module (Figure 12). Permuting the pathway similarity feature led to a 15% decrease in F1 score, while the number of indications for a disease (proxy of disease "popularity") and 6-th component in disease pathway embedding space caused decreases of approximately 3.5%. Other features, including various disease and drug pathway embeddings and molecular fingerprints, resulted in less than 3.5% decrease in F1 score.

The one-dimensional partial dependence plot (Figure A13) indicates that the steepest increase in success prediction occurs at a pathway similarity of around 0.25 and approximately 70 indications per disease. The two-dimensional partial dependence plot for pathway similarity and number of indications (Figure 13) shows that the model requires both high pathway similarity and a large number of indications to confidently predict a successful outcome.

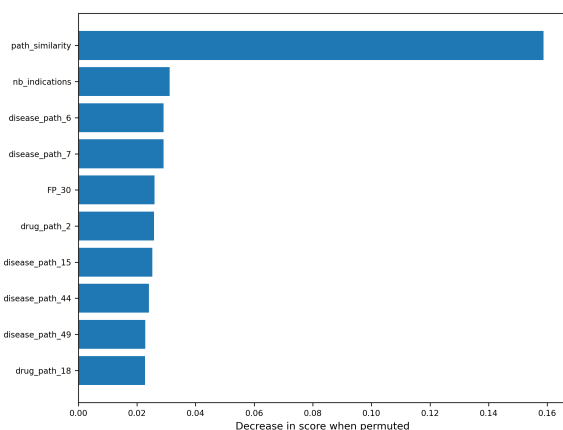


Figure 12: Permutation feature importance for trained XGBoost.

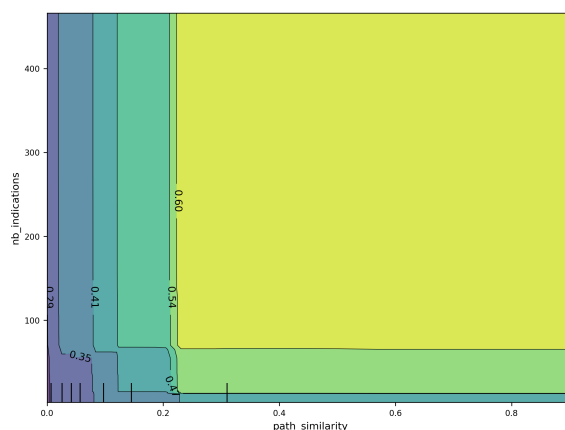


Figure 13: Two-dimensional partial dependence of the two most important features.

4.3 Failure and Success Cases

We added features to the test set indicating whether each observation is true positive (TP), true negative (TN), false positive (FP) or false negative (FN) and simply correctly classified (TP or TN). Pearson correlations between these confusion indicators and predictor features were computed, with caution: since the new variables are binary, these correlations should be interpreted as point-biserial correlations (Figures 14 and A14). We also examined the distributions of predictors for correctly versus incorrectly classified cases (Figures 15 and A15). Overall there are no major differences, and while some variables show slight biases, their interpretation remains speculative.

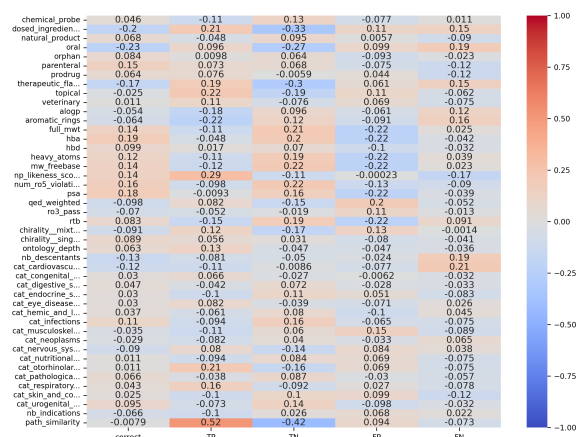


Figure 14: Correlation heatmap of XGBoost test classification results with predictors.

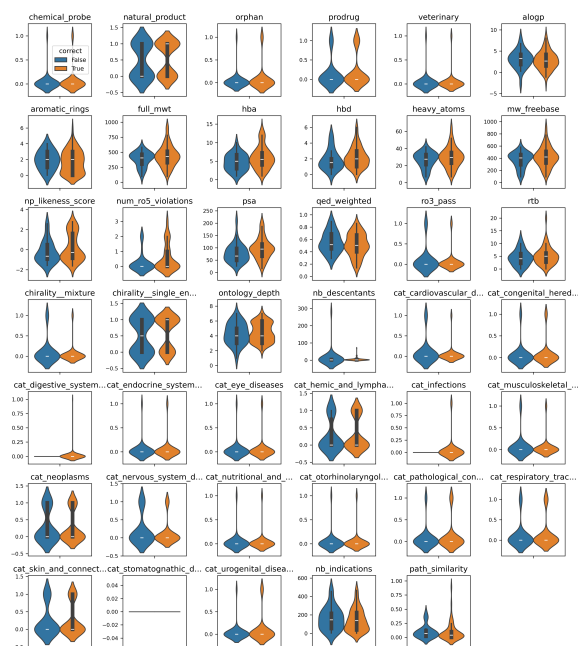


Figure 15: Feature distributions for samples correctly vs. incorrectly classified by XGBoost.

5 Discussion

The best-tuned model, XGBoost, showed moderate performance on the test data, with slightly higher precision (0.658) than recall (0.595). This indicates that the model produces more false negatives than false positives on this dataset. Since the primary goal is to prioritize new drug-disease indications, higher precision ensures that predictions labeled as positive are more reliable, while lower recall implies that some potentially promising combinations may be (systematically and unfairly) overlooked. Other models performed slightly worse, particularly kNN, suggesting that they struggled to capture non-linear patterns in noisy high-dimensional data or to handle class imbalance.

Heterogenous GNN was trained and tested separately and produced a large number of false positives, resulting in low precision of 0.429. This poor performance may be partially due to poor handling of class imbalance on our side, as we used a binary cross-entropy loss with positive weights proportional to the relative rarity of success labels, which may have been too stringent for the model. Further investigation is needed to identify proper strategies for handling class imbalance, as well as for hyperparameter and classification threshold tuning in GNNs. Other potential explanations for a poor model’s performance include the limited size of the training dataset, which may be insufficient for training such a complex model, as well as the need for strategies to prevent overfitting.

The two most important features identified by permutation importance are pathway similarity and number of indications for a disease, with higher values of both increasing the predicted probability of success. Pathway similarity is rather expected to be predictive, as it captures shared molecular mechanisms underlying drug action and disease pathogenesis. It is also unfortunately expected to see the number of indications being predictive of success since this feature reflects how intensively researched the disease is, this constitutes a bias towards well-studied diseases and potentially neglecting rare or

poorly studied ones. However, analysis of feature distributions for correctly and incorrectly classified cases didn't reveal any pronounced bias. This suggests that importance of number of indications likely reflects a bias in clinical research itself rather than a bias in the model. It can be also a reflection of data availability bias - rare diseases tend to be less annotated.

Another important source of bias is label assignment. Assessing the success of clinical trials is challenging due to poorly published results, and many drug-disease combinations have never been tested. Some combinations were not pursued because of clear mechanistic incompatibility, while others were limited by funding constraints. Even when a drug-disease indication is shown to be ineffective in a trial, it was typically preselected from thousands of potential alternatives as promising, otherwise, it would not have received funding. Furthermore, there is a publication bias reflected in better publication quality and documentation of successful trials, whereas some failed trials may be prematurely terminated without clear justification (e.g. citing funding issues while concealing drug inefficiency). One potential solution is to include "obvious negatives" combinations in the dataset or to employ semisupervised approaches, such as Positive-Unlabeled (PU) learning models, which rely on the positive class and treat the remaining data as unlabeled (mixture of positive and negative cases), allowing for indirect reconstruction of the negative class.

6 Conclusion

Although the evaluation results are likely biased toward better-studied diseases and demonstrate only moderate predictive performance, it is notable that a pathway similarity metric, based on simplified assumptions, combined with basic chemical and molecular descriptors can still provide meaningful predictions.

Future directions for ameliorating our approach would be:

- considering non-protein targets and biopolymer drugs,
- incorporating the directionality of associations with targets,
- expanding the scope of studied diseases,
- integrating labels from additional data sources,
- further exploring Graph Neural Networks to model molecular interaction networks, and
- going further in interpretation of pathway embeddings and molecular fingerprints by relating them back to the original interpretable pathways and chemical structures.

References

- [Bickerton et al., 2012] Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S., and Hopkins, A. L. (2012). Quantifying the chemical beauty of drugs. *Nat Chem*, 4(2):90–98.
- [Chen et al., 2025] Chen, J., Hu, Y., Cai, M., and et al. (2025). Trialbench: Multi-modal ai-ready datasets for clinical trial prediction. *Scientific Data*, 12:1564. Received 05 September 2024; Accepted 04 July 2025; Published 26 September 2025.
- [Ertl et al., 2008] Ertl, P., Roggo, S., and Schuffenhauer, A. (2008). Natural product-likeness score and its application for prioritization of compound libraries. *J. Chem. Inf. Model.*, 48(1):68–74. Epub 2007 Nov 23.
- [Gao et al., 2024] Gao, C., Pradeepkumar, J., Das, T., Thati, S., and Sun, J. (2024). Automatically labeling \$200b life-saving datasets: A large clinical trial outcome benchmark. arXiv preprint, submitted 13 June 2024, perpetual non-exclusive license.
- [Keramida et al., 2025] Keramida, P., Syrigos, N. K., Kouvela, M., Poulakou, G., Charpidou, A., and Fiste, O. (2025). Ai-driven drug repurposing: Applications and challenges. *Oncology Reports*. Open Access, published 13 November 2025, correspondence to Paraskevi Keramida.

A Appendix

A.1 Supplementary Tables

Table A1: Description of features retained for analysis.

Feature	Type	Description
Drug		
drug_id	str	Unique ChEMBL identifier
chemical_probe	bool	Well-characterized and selective, typically used in biological research
dosed_ingredient	bool	Active ingredient that has been dosed in humans
inorganic_flag	bool	Inorganic molecule
natural_product	bool	Derived from a natural source (not fully synthetic)
oral	bool	Known to be administered orally
orphan	bool	Intended for use against a rare condition
parenteral	bool	Known to be administered parenterally
prodrug	bool	Prodrug, activated after administration
therapeutic_flag	bool	Has a therapeutic application, opposed to e.g. imaging
topical	bool	Known to be administered topically
veterinary	bool	Has veterinary application as well
chirality	cat	achiral,single enantiomer, mixture,nan
alogp	float	Octanol-water partition coefficient, lipophilicity indicator
aromatic_rings	int	Number of aromatic rings
full_mwt	float	Molecular weight of the full compound including any salts
hba	float	Number of hydrogen bond acceptors
hbd	float	Number of hydrogen bond donors
heavy_atoms	int	Number of heavy (non-hydrogen) atoms
mw_freebase	float	Molecular weight of parent compound
np_likeness_score	float	Natural product likeness score [Ertl et al., 2008]
num_ro5_violations	int	Violations of Lipinski’s rule-of-five, using HBA and HBD definitions
psa	float	Polar surface area
qed_weighted	float	Weighted quantitative estimate of drug likeness [Bickerton et al., 2012]
ro3_pass	int	Passes the rule-of-three (mw < 300, logP < 3 etc)
rtb	int	Number of rotatable bonds
FP_#	float	First PCs in Morgan molecular fingerprint space
drug_path_#	float	First PCs in TF-IDF space derived from pathway lists
Disease		
disease_id	str	Unique EFO identifier
ontology_depth	int	Depth in disease classification tree
nb_descendants	int	Number of descendants in disease classifications tree
nb_indications	int	Number of drug indications
cat_#	bool	Ontological category other than immune system disorders
disease_path_#	float	First PCs in TF-IDF space derived from pathway lists
Drug–Disease		
success	bool	Consensus success or failure
path_similarity	float	Jaccard similarity of pathway lists
name_similarity	float	Cosine similarity of name embeddings in PubMedBERT (later excluded due to data leak)

Table A2: Best estimator parameters after hyperparameter tuning.

Model	Best parameters
XGBoost	colsample_bytree = 0.8, gamma = 0, learning_rate = 0.01, max_depth = 5, n_estimators = 300, subsample = 1.0
Logistic Regression	C = 0.1, l1_ratio = 0.25
Support Vector Machine	C = 1, degree = 2, gamma = scale, kernel = poly
Gradient Boost	learning_rate = 0.05, max_depth = 3, n_estimators = 100, subsample = 0.8
Random Forest	max_depth = 10, min_samples_leaf = 2, min_samples_split = 2, n_estimators = 100
k-Nearest Neighbors	n_neighbors = 5, p = 2, weights = distance

A.2 Supplementary Figures

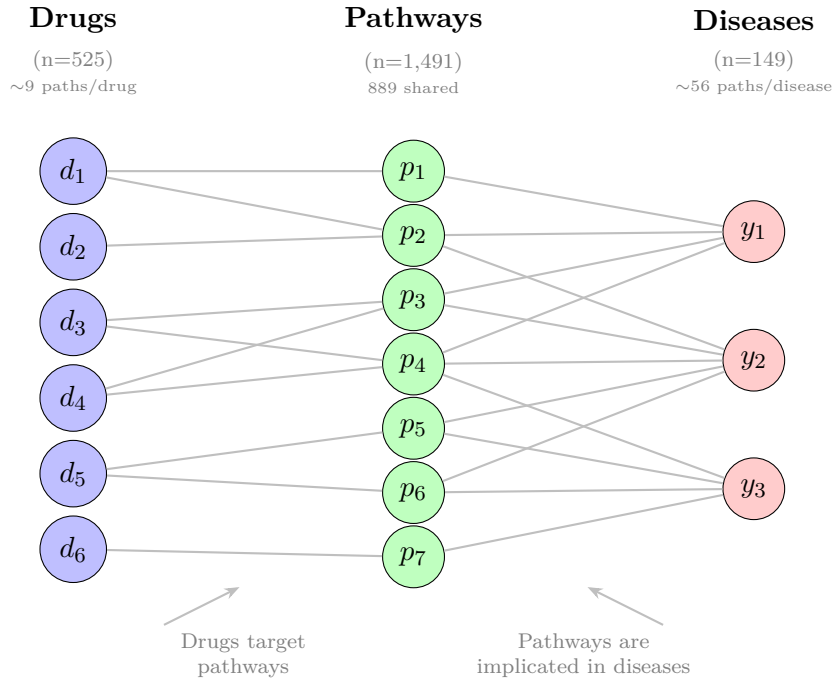


Figure A1: Tripartite graph representing the biological relationship between drugs, pathways, and diseases. Edges connecting drugs and diseases indicating clinical success are not represented.

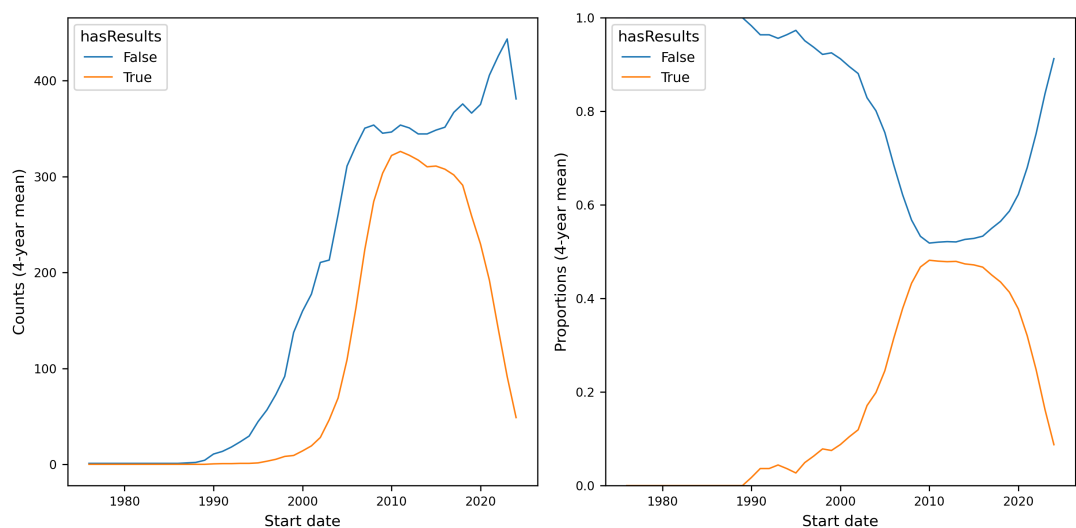


Figure A2: Counts and proportions of clinical trials with published results over time.

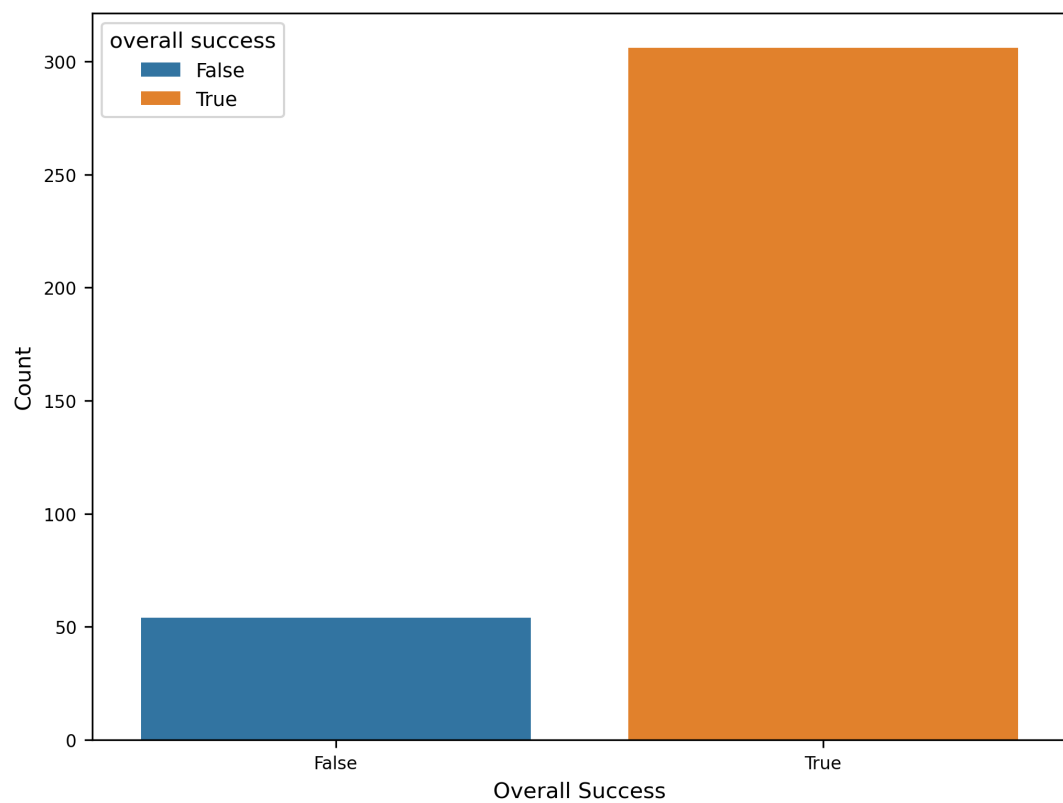


Figure A3: Distribution of success labels for drug-disease indications with missing start dates.

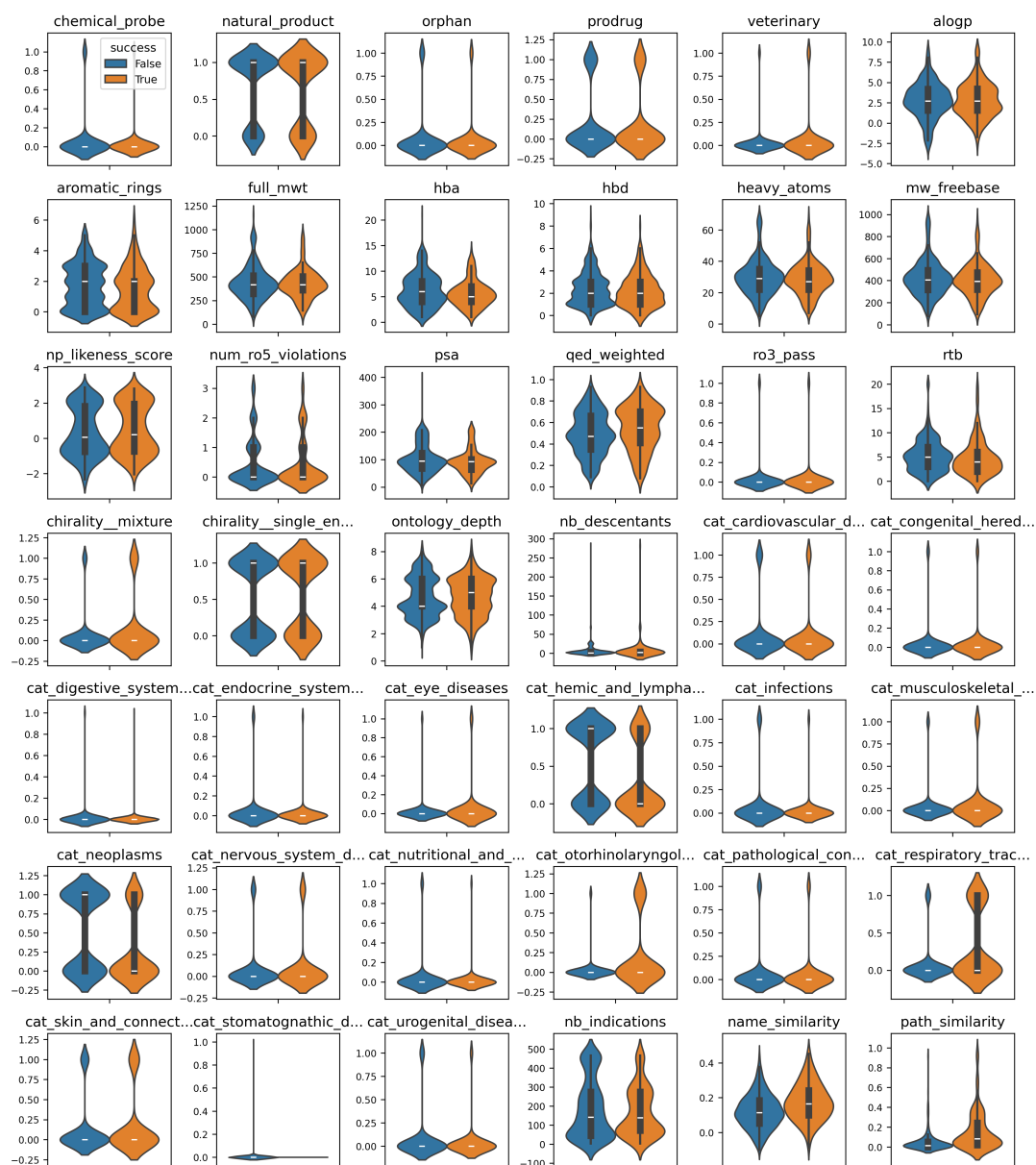


Figure A4: Feature distributions by success label.

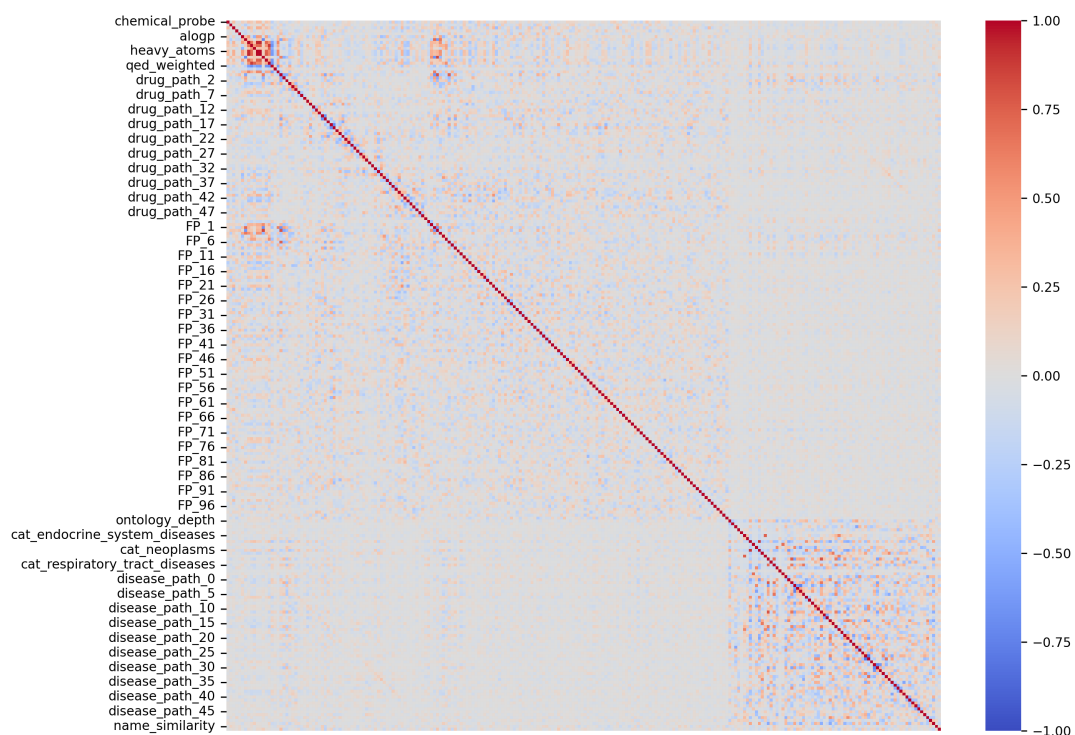


Figure A5: Correlation heatmap of all features.

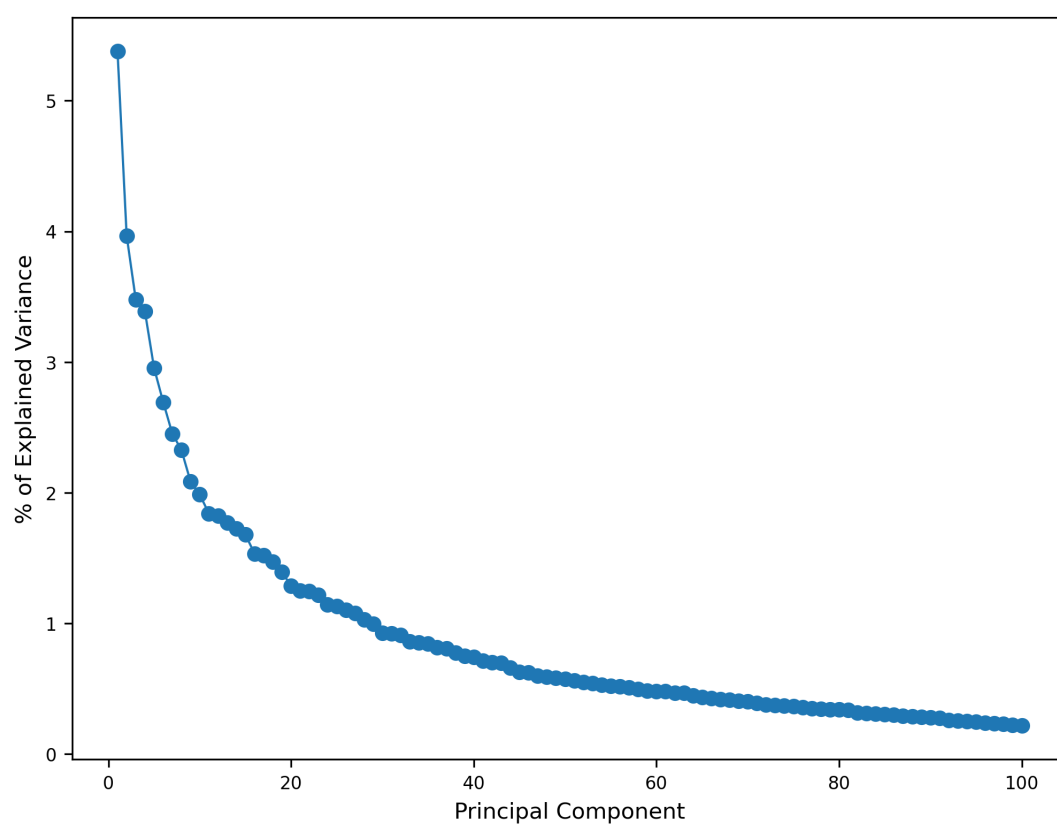


Figure A6: Scree plot of explained variance for PCA components.

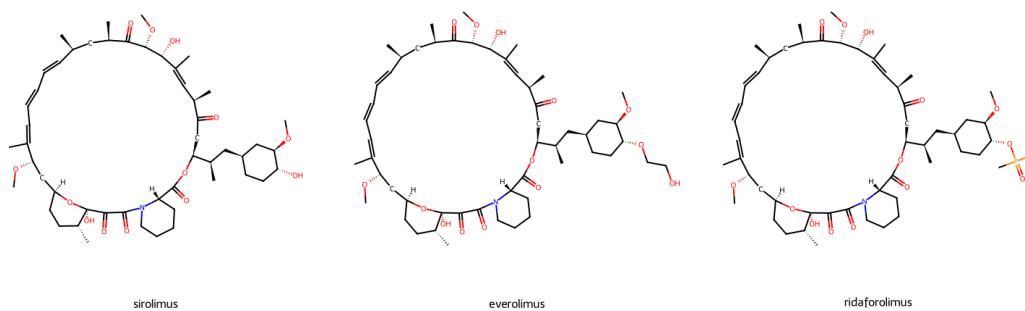


Figure A7: Structures of drugs corresponding to outlier indications.

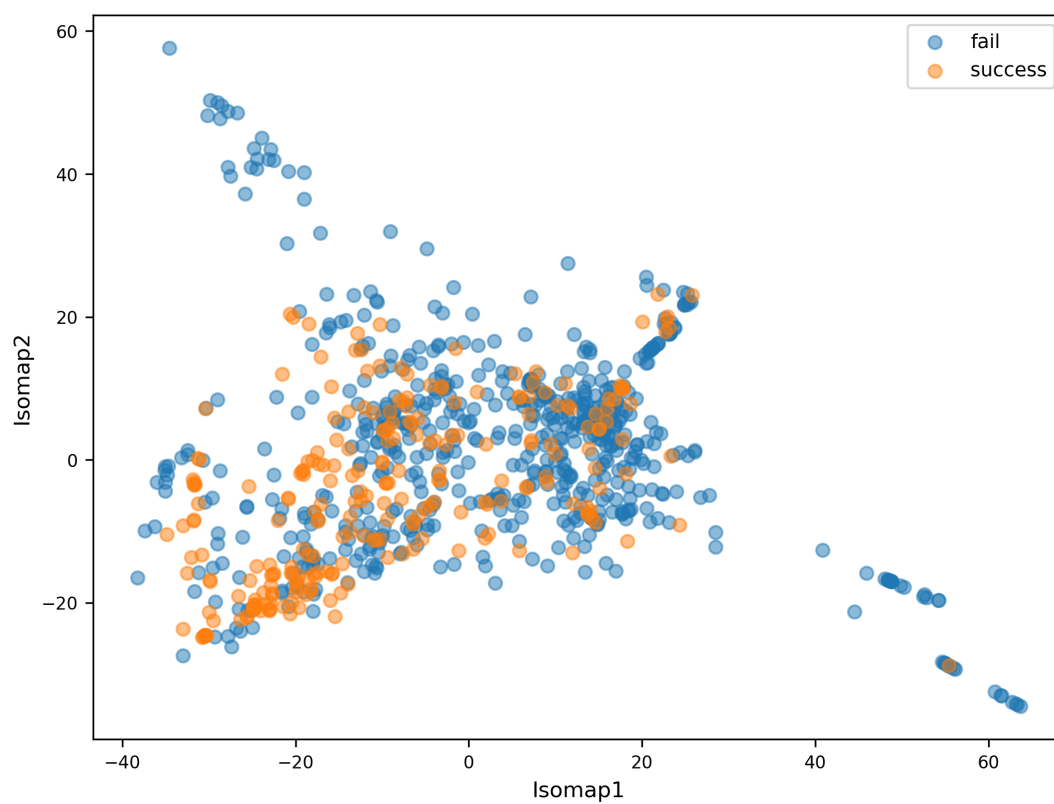


Figure A8: Isomap projection of samples in the feature space using 10 nearest neighbors.

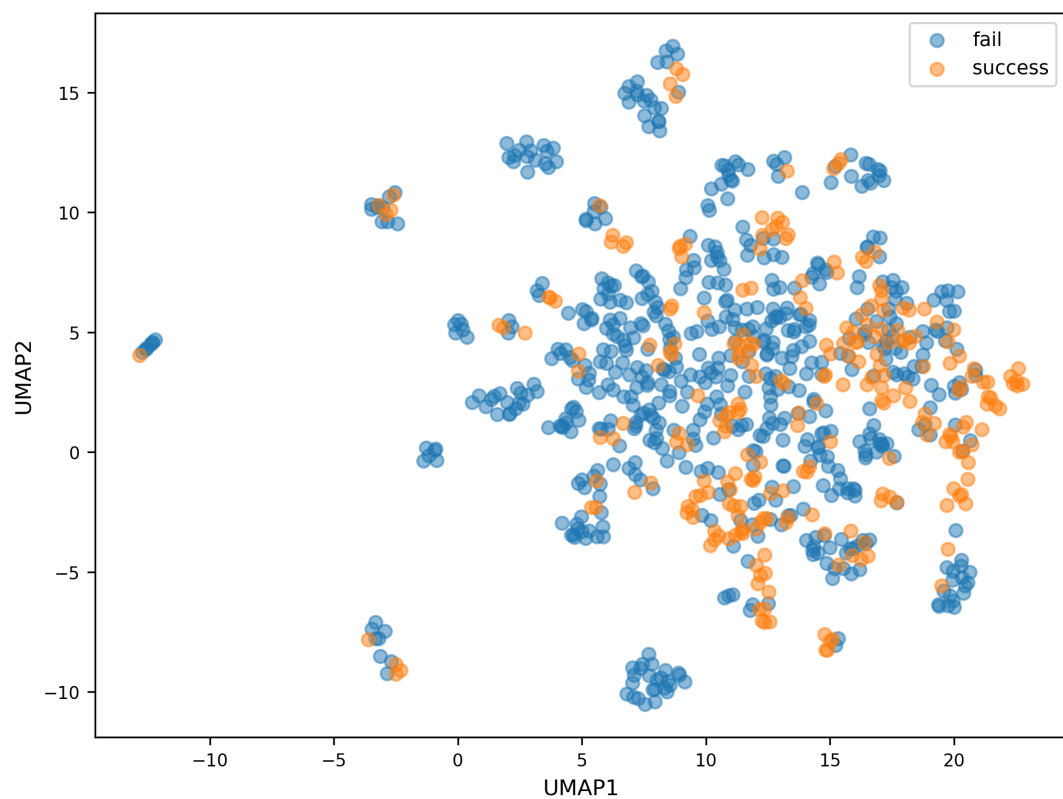


Figure A9: UMAP projection of samples in the feature space using 7 nearest neighbors and a minimal distance of 1.

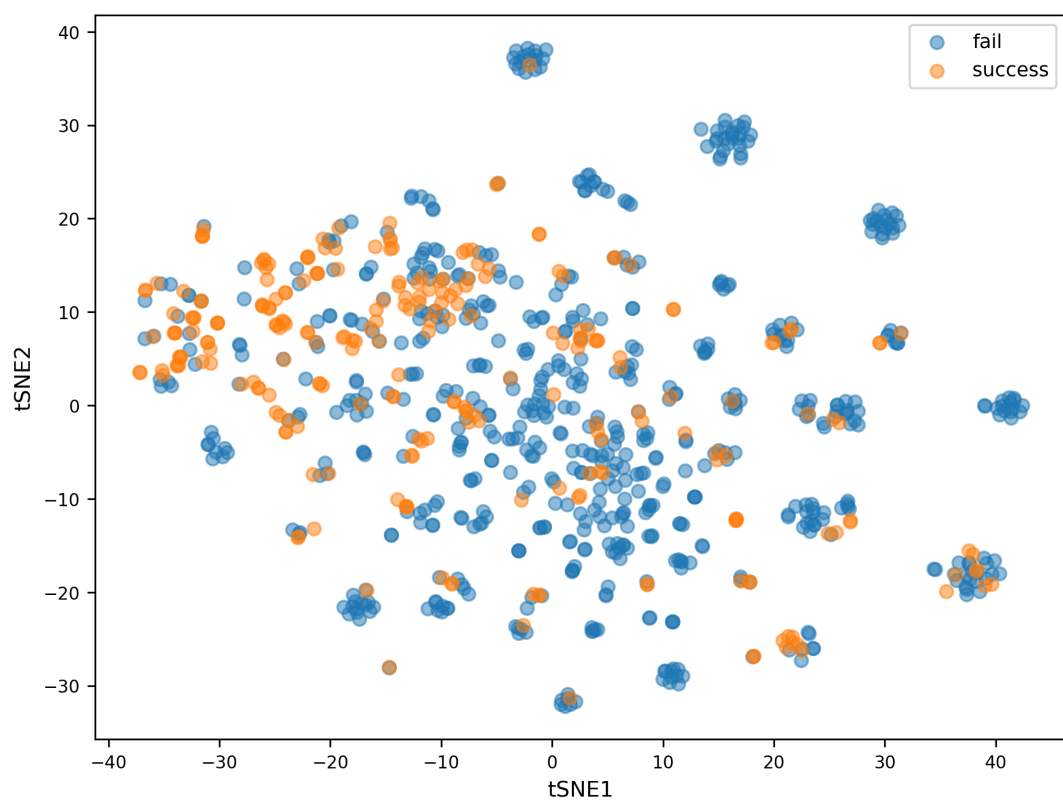


Figure A10: t-SNE projection of samples in the feature space using a perplexity of 40.

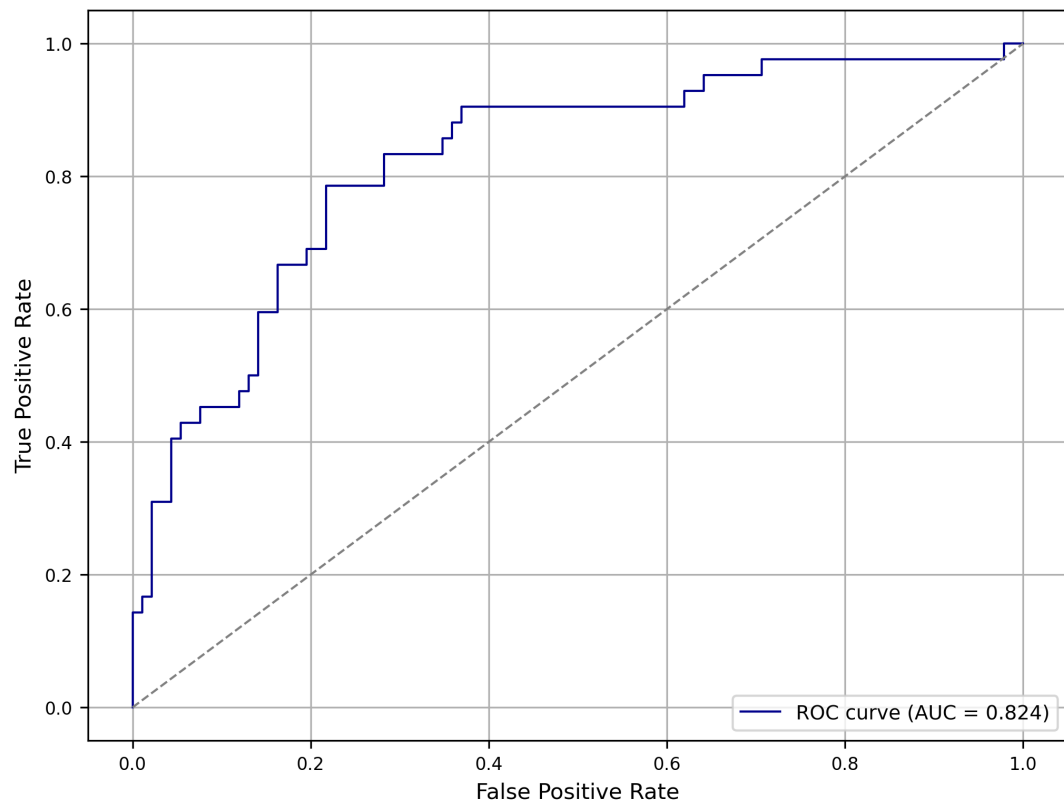


Figure A11: ROC curve for XGBoost on the test dataset.

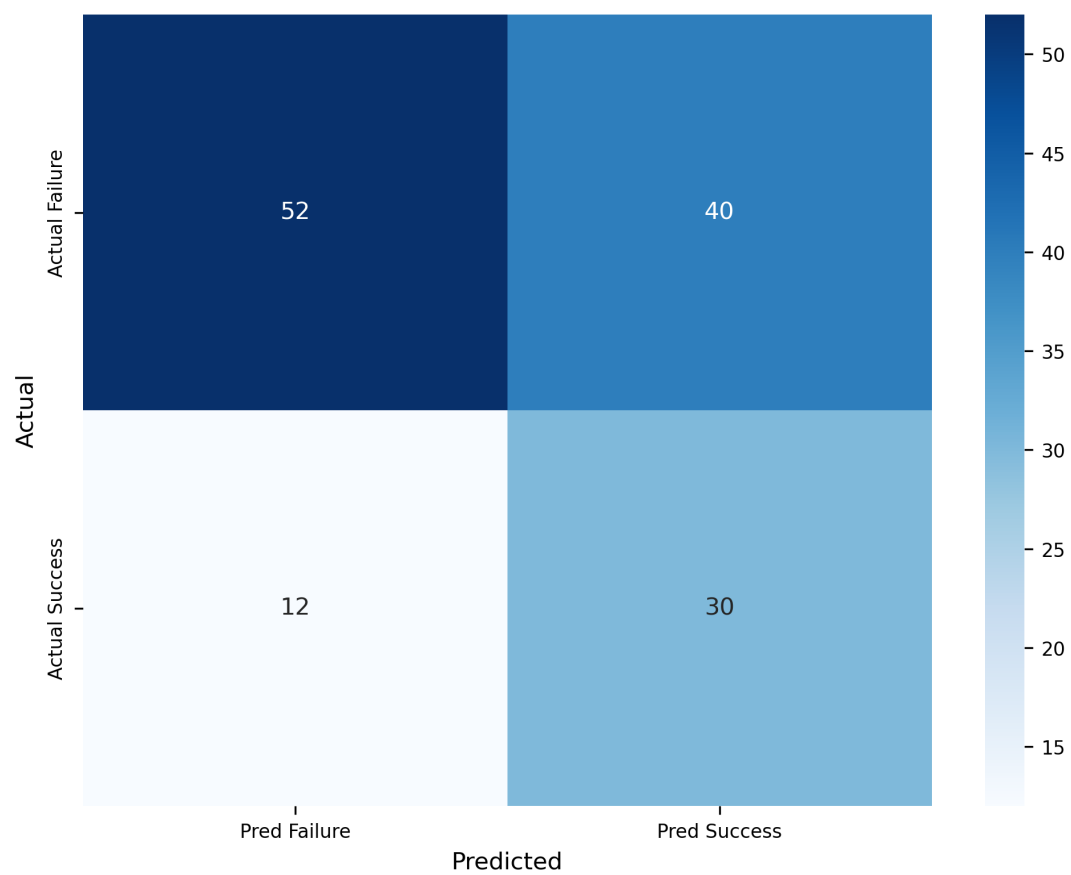


Figure A12: Confusion matrix for Heterogenous GNN on the test dataset.

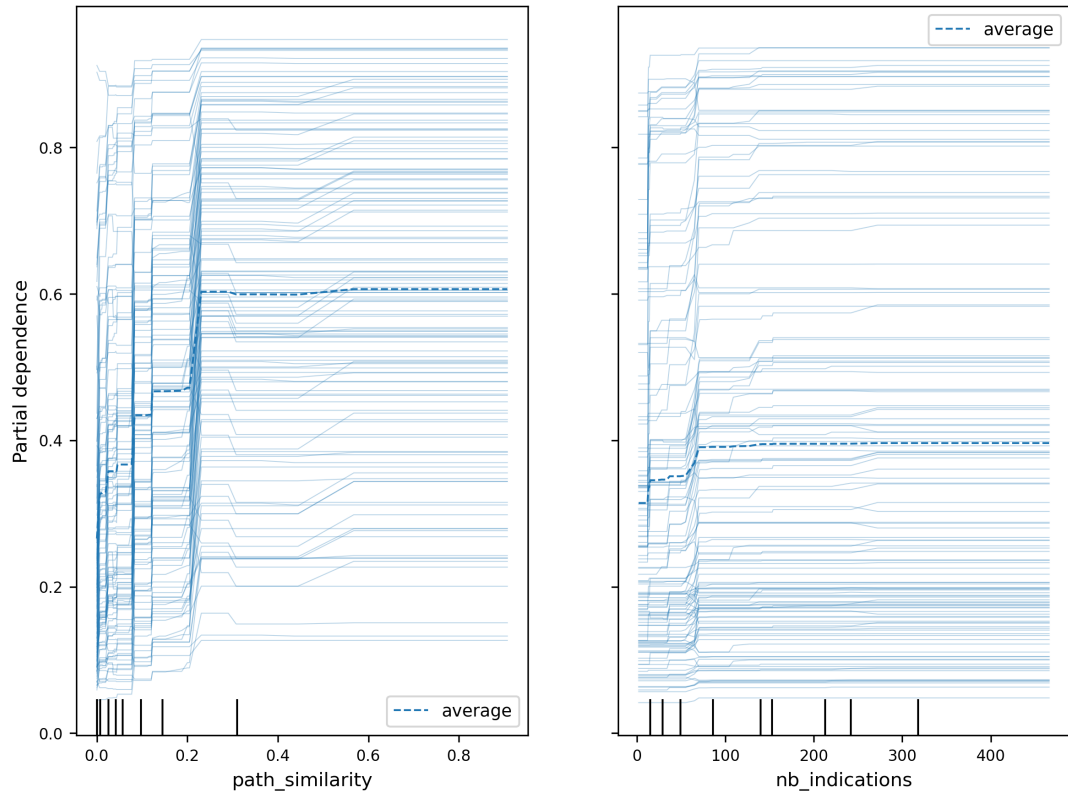


Figure A13: Partial dependence of the two most important features.

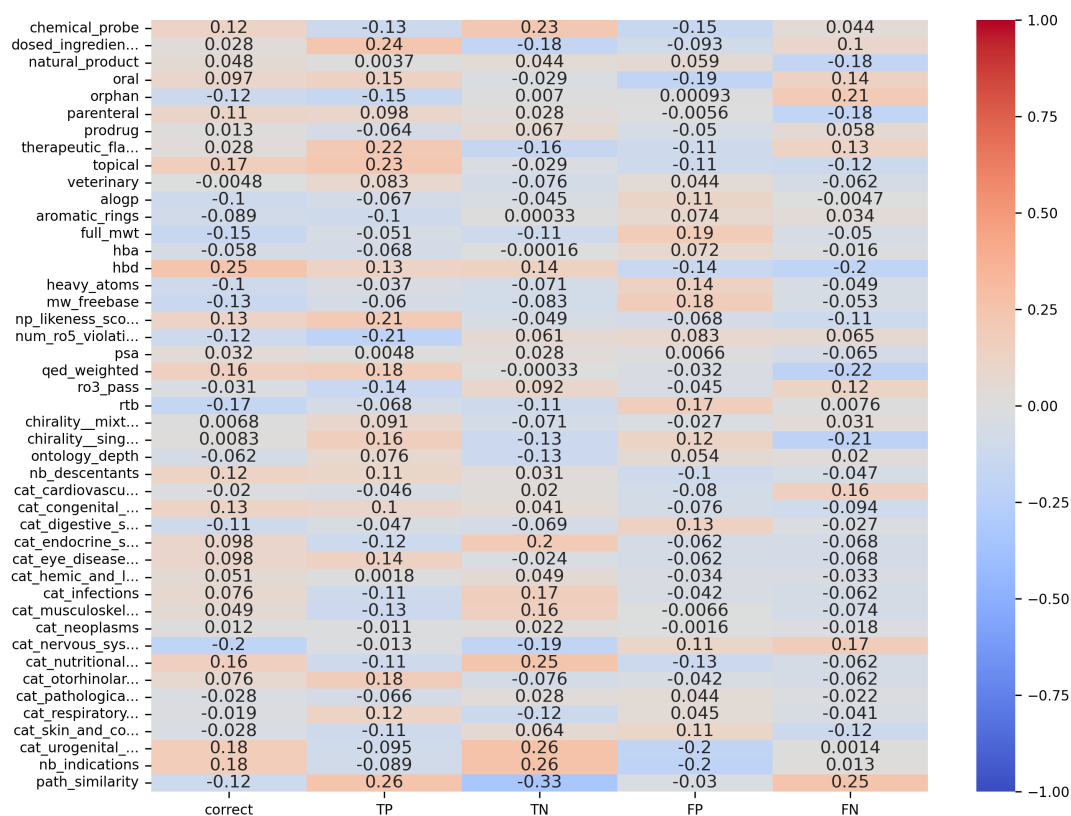


Figure A14: Correlation heatmap of Heterogenous GNN test classification results with predictors.

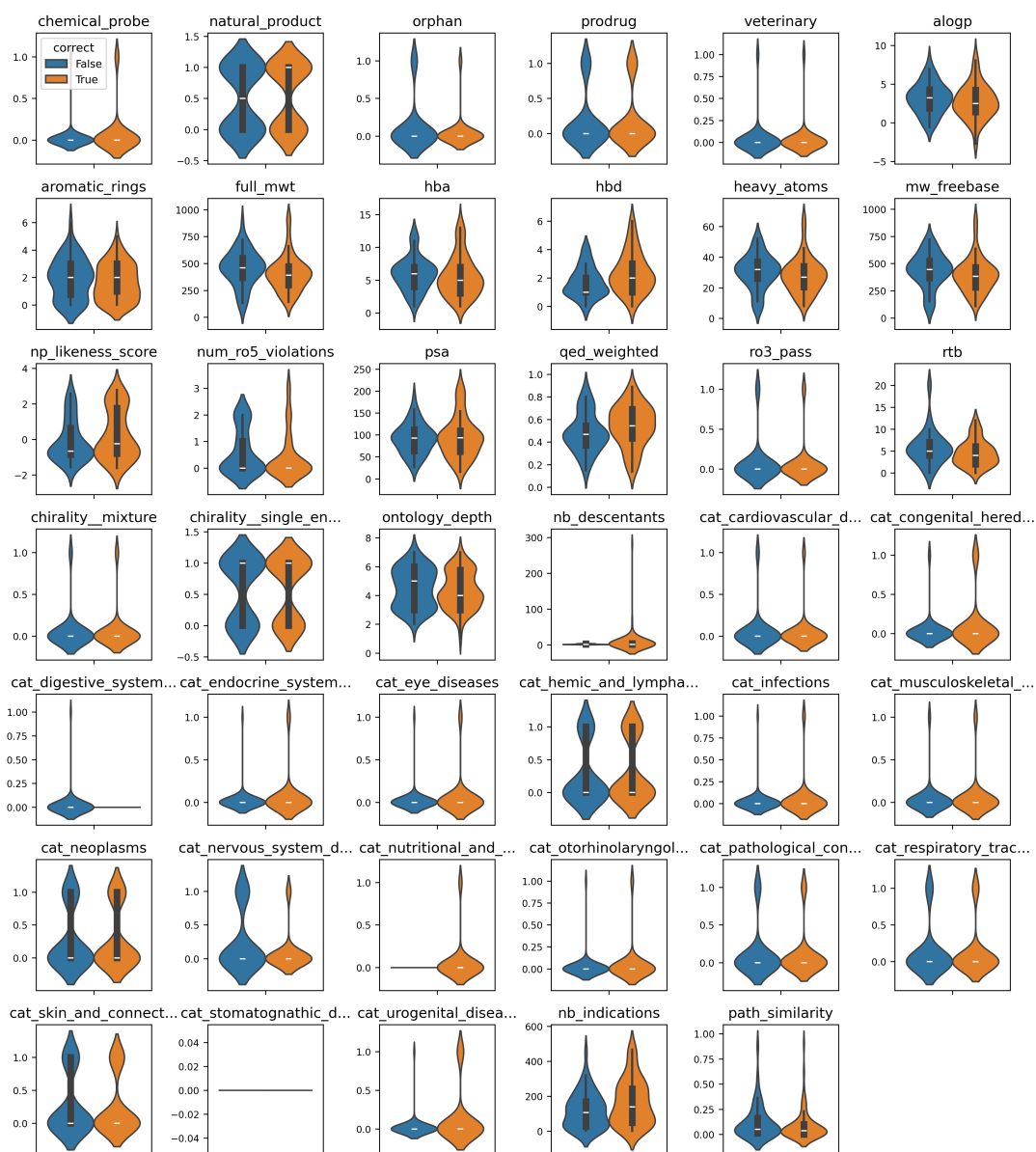


Figure A15: Feature distributions for samples correctly vs. incorrectly classified by Heterogenous GNN.

A.3 Supplementary Algorithms and Listings

Algorithm A1 Drug–Disease Success Labeling

Require: Clinical trial data for a drug–disease indication

Ensure: Success, fail or unknown label

```
if phase 4 is passed then
  label  $\leftarrow$  success
else if any clinical trial exists and has results then
  Compute median p-value of trials
  if median p-value  $< 0.05$  then
    trial label  $\leftarrow$  success
  else if median p-value  $> 0.2$  then
    trial label  $\leftarrow$  fail
  else
    trial label  $\leftarrow$  unknown
  end if
  Let  $P_{\max}$  be the highest phase tried
  Let  $S$  and  $F$  be the number of successes and fails in phase  $P_{\max}$ 
  if  $P_{\max} \geq 3$  and  $S \geq 2 \times F$  then
    label  $\leftarrow$  success
  else if  $F \geq 2 \times S$  then
    label  $\leftarrow$  fail
  else
    label  $\leftarrow$  unknown
  end if
end if
if label is still unknown then
  Let  $D_{\text{latest}}$  be the latest trial date
  if  $D_{\text{latest}} < 2020$  then
    label  $\leftarrow$  fail
  else
    label  $\leftarrow$  unknown
  end if
end if
return label
```

Listing 1: HeteroData graph structure

```
HeteroData(
  drug={ x=[525, 127] },
  disease={ x=[149, 25] },
  pathway={ x=[1487, 1] },
  (drug, interacts, pathway)={ edge_index=[2, 17338] },
  (disease, associated_with, pathway)={ edge_index=[2, 12064] },
  (drug, tried_against, disease)={
    edge_index=[2, 890],
    edge_label=[890],
  })
```